

120 Days of Quantitative Finance

Week 16: Stochastic Calculus + Time Series

Day 112: Stochastic Volatility vs Constant Volatility

1 Introduction

Option pricing models differ significantly in how they treat volatility.

This section compares constant volatility models (Black-Scholes) with stochastic volatility models (Heston).

2 Black-Scholes Model

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

- Volatility σ is constant
- Closed-form solution exists

3 Limitations of Constant Volatility

- Cannot capture volatility clustering
- Fails to explain volatility smile
- Assumes lognormal returns

4 Heston Model

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^{(1)}$$

$$dv_t = \kappa(\theta - v_t)dt + \xi \sqrt{v_t} dW_t^{(2)}$$

5 Key Differences

- Volatility is stochastic and time-varying
- Captures mean-reversion in variance
- Includes correlation between price and volatility

6 Volatility Smile

- Black-Scholes assumes flat implied volatility
- Heston produces realistic volatility smiles

7 Leverage Effect

- Negative correlation between returns and volatility
- Captured by Heston via ρ

8 Comparison Summary

- Black-Scholes: simple, tractable, unrealistic
- Heston: complex, realistic, computationally intensive

9 Financial Implications

- Constant volatility underprices tail risk
- Stochastic volatility improves pricing accuracy

10 Conclusion

Stochastic volatility models provide a more realistic framework for option pricing, especially in capturing market phenomena like volatility smiles and clustering.